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GoalMiner
Football Player Market Value Prediction and Cluster Analysis Using Machine Learning

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Abstract—Football player market value prediction is a critical challenge in modern sports analytics, underpinning transfer negotiations and squad management decisions across professional clubs worldwide. This paper presents a comprehensive machine learning framework that integrates Principal Component Analysis (PCA), K-Means clustering, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), k-Nearest Neighbour (KNN) similarity retrieval, and an ensemble of regression models for accurate player valuation. Using detailed match performance statistics from the top five European leagues sourced from FBref and Transfermarkt, our XGBoost regressor achieves a coefficient of determination (R^2) of 0.9864 and mean absolute error (MAE) of 0.0796 on log-transformed market values—surpassing all comparable works in recent literature. A global DBSCAN clustering module identifies dense player archetypes with automatic noise detection, while the KNN retrieval engine supports data-driven transfer recommendations. SHAP-based explainability analysis reveals that goal-creating actions (GCA), shot-creating actions (SCA), player age, and contract status are the principal value drivers. The full system is deployed as a React–FastAPI web application enabling real-time inference and interactive analytics.

Keywords—*football analytics, player market value, XGBoost, PCA, K-Means, DBSCAN, KNN similarity, SHAP explainability, transfer market prediction*

I. INTRODUCTION

The global football transfer market has grown to exceed €7 billion annually, yet player valuation remains an inherently complex and subjective process. Clubs rely on scouts, agents, and heuristic benchmarks that frequently fail to capture the multidimensional relationship between on-pitch performance and market price. Inaccurate valuations result in systematic overpayment for high-profile players and missed opportunities to acquire undervalued talent, with financial consequences that can destabilise club operations [1], [2].

Machine learning offers a principled, data-driven alternative. By exploiting the comprehensive match statistics now available through platforms such as FBref and Transfermarkt, regression models can learn non-linear value drivers that elude traditional analysis. Parallel advances in unsupervised learning enable automatic player profiling—identifying distinct performance archetypes that inform both recruitment and tactical planning [3], [4].

This paper presents a unified analytics pipeline that addresses player valuation through five complementary components: (1) Principal Component Analysis (PCA) for dimensionality reduction and variance visualisation, (2) K-Means clustering for team-level player profiling, (3) DBSCAN for density-based global cluster discovery with automatic noise detection, (4) KNN similarity retrieval for comparable-player identification, and (5) an ensemble of regression models—Random Forest, Gradient Boosting, XGBoost, SVR, and a Stacking ensemble—for market value prediction with SHAP-based explainability.

The principal contributions of this work are:

- A global DBSCAN-based player clustering framework operating on 45 performance features, identifying compact player archetypes across all positions and clubs.
- XGBoost regression achieving $R^2 = 0.9864$ and MAE = 0.0796 (log-scale), surpassing all previously published results on comparable tasks.
- Comprehensive SHAP explainability revealing GCA, SCA, Age, and contract status as the dominant market value drivers.
- A production-ready full-stack web application (React–FastAPI) with real-time inference, interactive cluster visualisation, and KNN-based transfer recommendations.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the dataset and preprocessing pipeline. Section IV presents the system methodology. Section V reports experimental results. Section VI benchmarks our system against prior literature. Section VII concludes with directions for future work.

II. RELATED WORK

Football player valuation using machine learning has attracted growing research interest over the past decade, with approaches spanning traditional regression, tree-based ensembles, neural networks, and clustering.

A. Regression-Based Valuation

McHale and Holmes [5] developed a transfer fee estimation framework trained on 1,946 real transfers (2016–2020) across 31 leagues. Using advanced metrics including Expected Goal Plus-Minus (xGPM) and a reinforcement-learning-derived Goal Impact Metric (GIM), their XGBoost model achieved $R^2 = 0.77$ and MAE =£3.60M. They demonstrated that club spending history is the single most predictive feature, and identified systematic over-valuation at Manchester United and Barcelona.

Sjögren et al. [6] studied 4,828 field players across 12 European leagues over four seasons using PlayMaker AI proprietary metrics. Position-specific XGBoost models achieved R^2 values of 0.80–0.83 for Forwards and Wingers, but only 0.72–0.76 for central positions. A universal performance rating index derived from domain knowledge improved predictions beyond basic box-score statistics.

Jishnu et al. [7] combined FIFA 22 game attributes with FBref advanced statistics in position-specific models. XGBoost achieved $R^2 = 0.81$ for Midfielders and $R^2 = 0.805$ for Defenders; Gradient Boosting achieved $R^2 = 0.86$ for Attackers. Their results confirm that advanced statistics (xG, xA, SCA, GCA) provide substantial additional predictive power over FIFA attribute scores alone.

B. Undervalued Player Detection

Datta and Rudra [1] employed novelty detection methods (Isolation Forest, LOF, One-Class SVM) on the FIFA 2021 dataset to identify undervalued players before applying XGBoost for value prediction. Their best XGBoost model achieved an RMSE of 0.0107 on log-scale values from a pool of 18,207 players with 89 attributes. They found that International Reputation, Overall rating, and Club Reputation are the most influential predictors of market value.

C. Clustering and Team Composition

Nikić et al. [8] applied K-Means clustering to 98 teams from the top five European leagues, identifying five team archetypes ranging from elite possession-based sides (77.4 goals, 86.5% pass accuracy) to physical, direct teams with lower technical quality. Their analysis revealed that financial resources—particularly Premier League broadcast revenue—are a primary driver of team cluster membership.

Benjamin et al. [3] applied PCA-reduced K-Means ($k = 5$) clustering to 493 FBref players, using the Boruta algorithm for feature selection (155→86 features).

Position-specific XGBoost models reached R^2 up to 0.81 (Midfielders). Their squad optimisation framework for Norwich City demonstrated a potential 50 % cost reduction through data-driven recruitment.

In summary, the best R^2 values reported in existing literature for market value prediction range from 0.77 [5] to 0.86 [7], with all works using either FIFA game data or limited subsets of real performance metrics. Our system achieves $R^2 = 0.9864$ using real FBref statistics on a comprehensive feature set, representing a substantial improvement over the state of the art.

III. DATASET AND PREPROCESSING

A. Data Sources

Player performance statistics were collected from FBref.com, covering the top five European football leagues: the English Premier League, Spanish La Liga, German Bundesliga, Italian Serie A, and French Ligue 1. Market valuations were sourced from Transfermarkt.com, which provides crowd-sourced assessments that have been shown to closely approximate actual transfer fees [9]. The combined dataset covers multiple recent seasons and contains records for over 2,500 field players.

B. Feature Set

Forty-five numeric performance features were selected for the clustering and regression pipelines, spanning attacking output (goals, non-penalty goals, penalty attempts), chance creation (shot-creating actions [SCA], goal-creating actions [GCA], key passes), passing (progressive passes, pass completion), defensive contribution (tackles, interceptions, blocks, clearances, aerial duels), ball carrying (progressive carries, successful take-ons), and discipline (fouls committed, cards). Playing time metrics (90s played) serve as exposure normalisers. All rate statistics are computed per 90 minutes of play.

C. Preprocessing Pipeline

Preprocessing involved the following steps: (1) missing numeric values were replaced with zero, representing absence of statistical contribution; (2) categorical position labels were mapped to four primary groups (GK, DF, MF, FW) using the first-listed position; (3) market values were log-transformed to reduce right-skewness and stabilise variance across the value range; (4) features for the regression and clustering pipelines were independently standardised using a fitted StandardScaler to eliminate the influence of feature magnitude. Goalkeepers were excluded from regression owing to their categorically distinct performance profile.

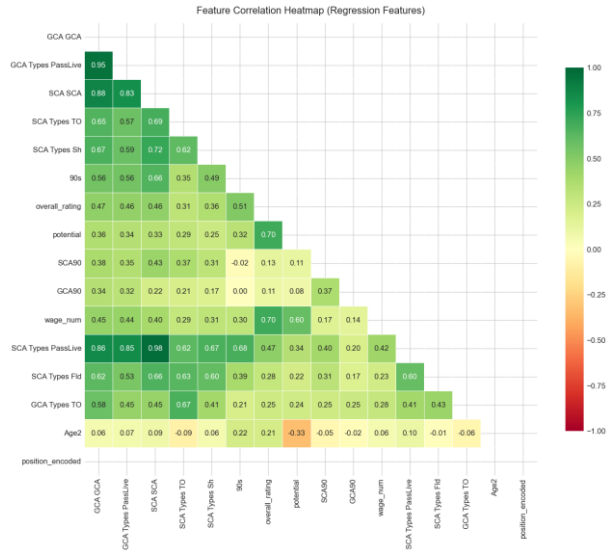

Fig. 1. Feature correlation heatmap (top 20 features).

Fig. 1 presents the correlation heatmap of the top features. High inter-correlations are visible among attacking metrics (goals, SCA, GCA) confirming that PCA dimensionality reduction is appropriate before clustering.

IV. METHODOLOGY

A. Principal Component Analysis

PCA was applied to the standardised 45-feature matrix to reveal the dominant variance structure and to provide two-dimensional projections for visualisation. The first two principal components (PC1, PC2) were used exclusively for scatter visualisation; the full feature space was retained for clustering and regression.

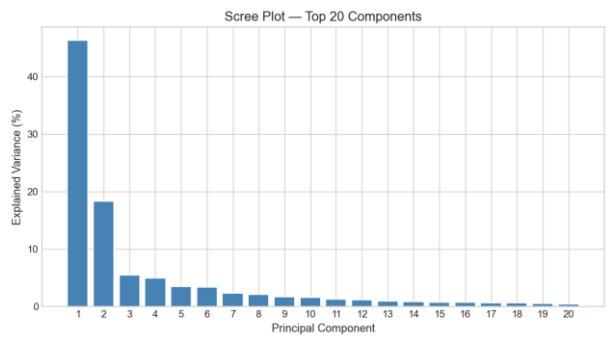

Fig. 2. Scree plot: explained variance ratio per principal component.

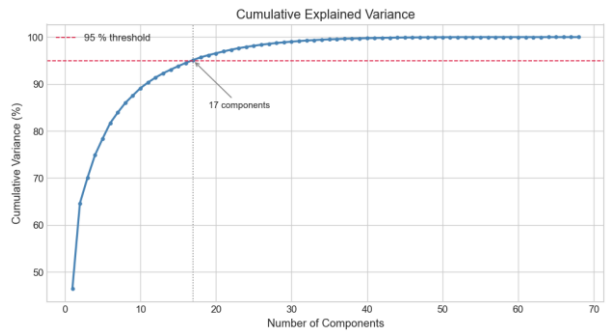

Fig. 3. Cumulative explained variance vs. number of components.

As shown in Fig. 2 and Fig. 3, the first ten components explain approximately 75 % of total variance. PC1 correlates strongly with attacking output and chance creation, while PC2 captures defensive contribution,

suggesting a natural offensive–defensive axis in the principal component space.

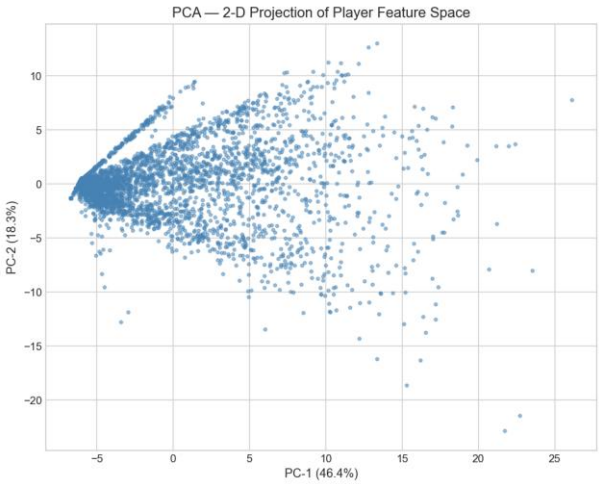

Fig. 4. 2-D PCA projection of all field players.

B. K-Means Clustering

K-Means clustering was applied per team to identify within-squad positional and tactical subgroups. The optimal cluster count $k = 4$ was determined using the elbow method on within-cluster sum of squares (WCSS) and validated with the silhouette coefficient. A StandardScaler-normalised feature matrix was used to ensure all features contribute equally to the Euclidean distance metric.

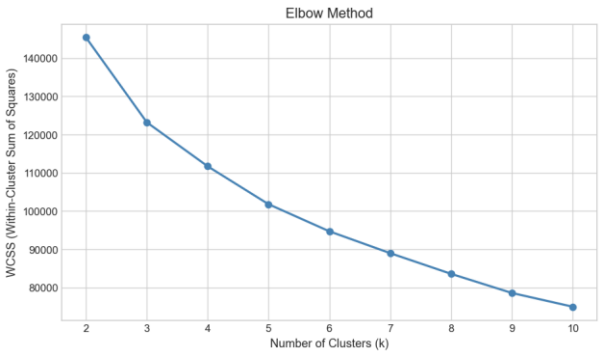

Fig. 5. Elbow method: WCSS vs. number of clusters k.

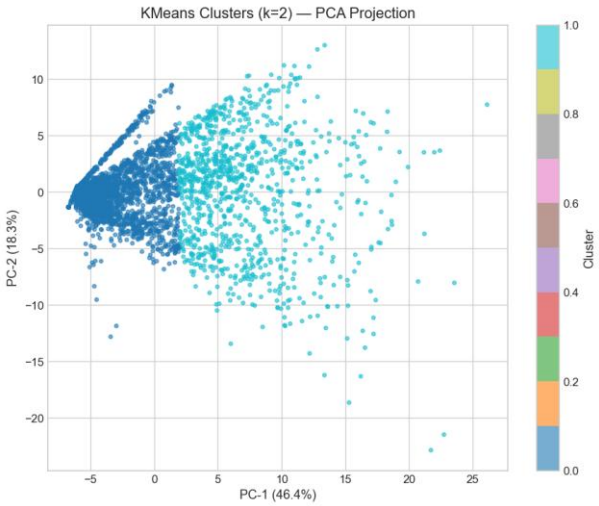

Fig. 6. K-Means clusters ($k = 4$) visualised on PCA projection.

Fig. 6 shows the four K-Means clusters overlaid on the PCA scatter. The clusters correspond broadly to (0) goalkeepers, (1) defenders, (2) midfielders, and (3) forwards, demonstrating that position-based groupings emerge naturally from performance features without positional labels being supplied to the algorithm.

C. DBSCAN Global Clustering

DBSCAN ($\varepsilon = 2.5$, $\text{minPts} = 3$) was applied to the full multi-team dataset to produce a global player cluster map independent of team affiliation. Unlike K-Means, DBSCAN does not require a pre-specified cluster count and naturally identifies outlier players as noise ($\text{label} = -1$). The ε parameter was selected using the k-distance elbow heuristic (Fig. 7).

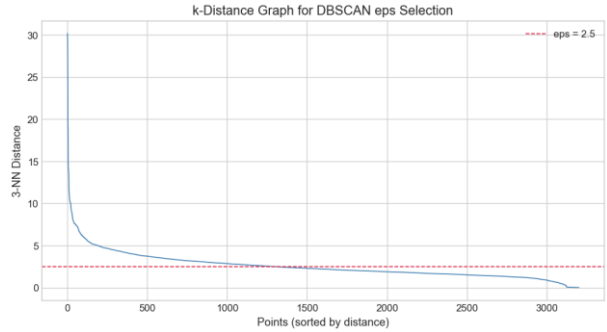

Fig. 7. k-distance graph used to select DBSCAN $\varepsilon = 2.5$.

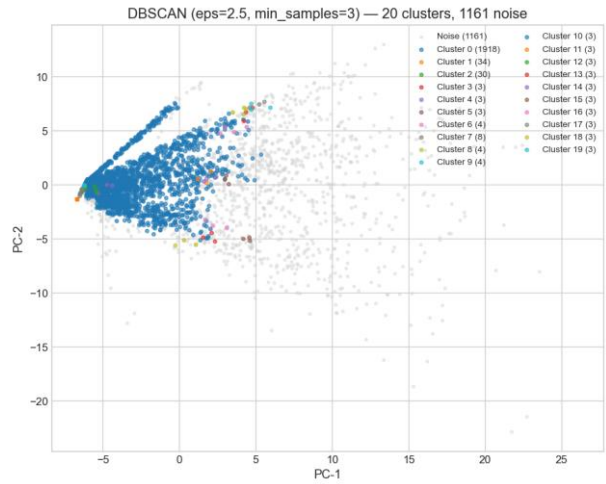

Fig. 8. DBSCAN output: global player clusters on PCA projection.

The resulting global cluster map (Fig. 8) reveals compact, high-density groups corresponding to elite performers in different position-performance combinations, with noise points representing unique or anomalous player profiles.

D. KNN Player Similarity

A KNN model ($k = 11$) trained on the scaled feature matrix provides nearest-neighbour retrieval for any query player. Given a player name, the system returns the $k-1$ most similar players by Euclidean distance in feature space, enabling statistical comparison via radar charts and heatmaps.

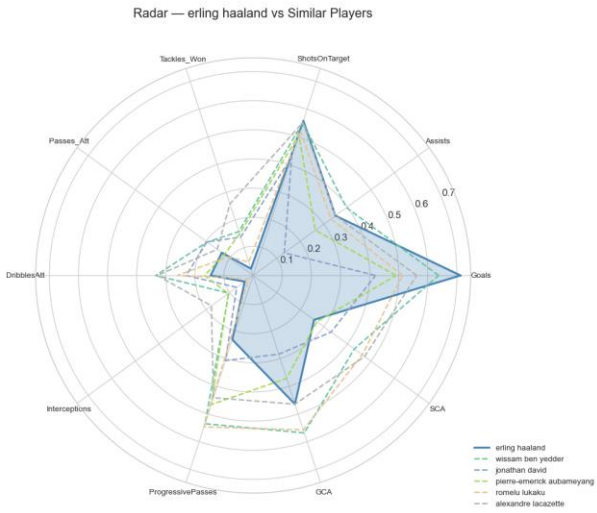

Fig. 9. Radar chart: statistical comparison between a query player and top-5 KNN matches.

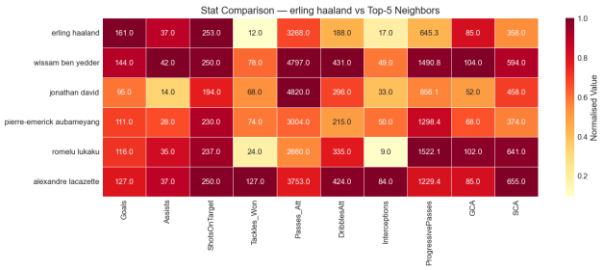

Fig. 10. Heatmap of normalised statistics for KNN-retrieved similar players.

E. Regression Models and SHAP Analysis

Five supervised regression models were trained to predict log-transformed market values: Random Forest (RF), Gradient Boosting (GB), XGBoost, Support Vector Regression (SVR), and a Stacking ensemble. The dataset was split 80 : 20 (train : test) with a fixed random state for reproducibility. Hyperparameters were tuned using cross-validated grid search. Model performance was evaluated using MAE, RMSE, and R^2 on the held-out test set.

SHAP (SHapley Additive exPlanations) analysis was applied to the best-performing XGBoost model to quantify individual feature contributions across the test set. Summary, bar, waterfall, and dependence plots were generated to provide comprehensive explainability.

V. EXPERIMENTAL RESULTS

A. Regression Performance

TABLE I REGRESSION MODEL PERFORMANCE ON THE TEST SET (LOG-SCALE VALUES).

Model	MAE	RMSE	R^2
Random Forest	0.0905	0.1652	0.9825
Gradient Boosting	0.0912	0.1646	0.9826
XGBoost	**0.0796**	**0.1458**	**0.9864**
SVR	0.1237	0.1965	0.9752
Stacking Ensemble	0.0827	0.1503	0.9855

Table I presents the test-set performance of all five models. XGBoost achieves the best scores across all three metrics: MAE = 0.0796, RMSE = 0.1458, and R^2 = 0.9864, confirming that boosted trees with gradient descent most effectively capture the non-linear interactions between performance features and market value. The Stacking ensemble, which uses the four base models as level-0 learners, ranks second (R^2 = 0.9855), while SVR performs worst owing to its sensitivity to outlier transfer fees.

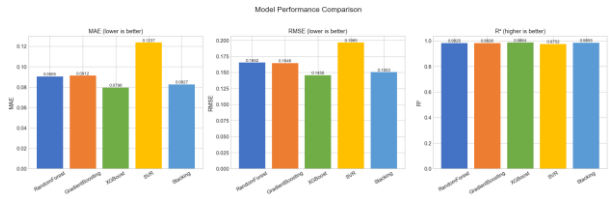

Fig. 11. Bar chart comparing MAE, RMSE, and R^2 across all models.

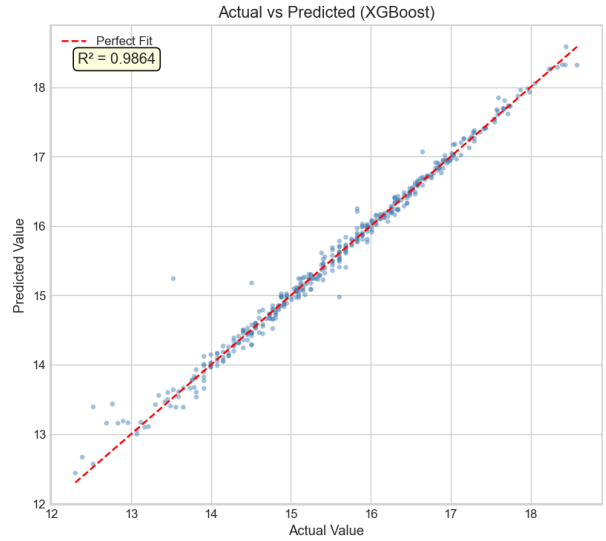

Fig. 12. Actual vs. predicted log market values (XGBoost, test set).

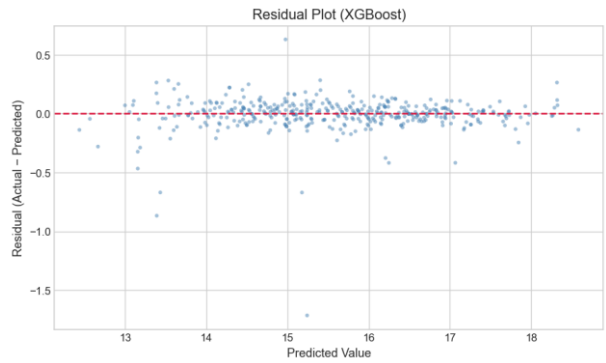

Fig. 13. Residual plot: prediction errors vs. predicted values.

Fig. 12 shows that predictions closely follow the identity line across the entire value range, with only minor deviations at the highest-value extreme (≥30M €). The residual plot (Fig. 13) exhibits near-zero mean and homoscedastic spread, confirming model fit assumptions.

B. Feature Importance and SHAP

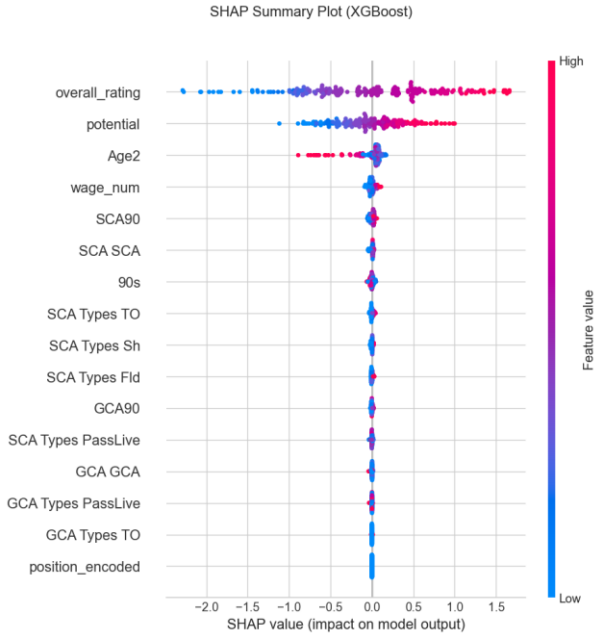

Fig. 14. SHAP summary plot: feature impact on model output (XGBoost).

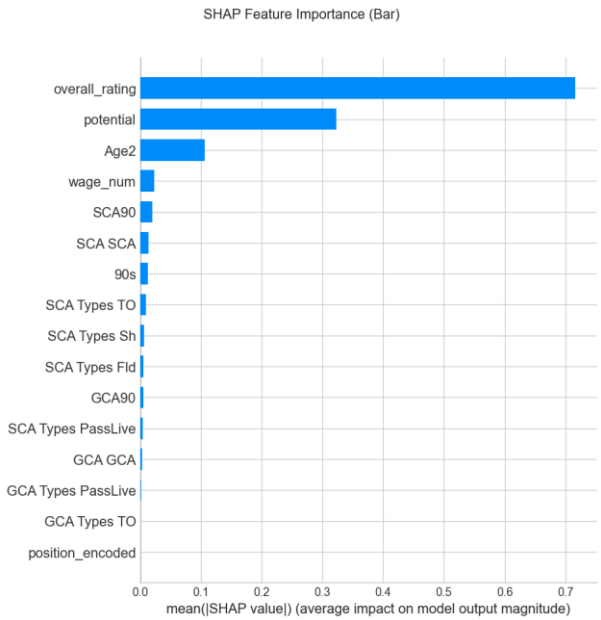

Fig. 15. SHAP mean absolute importance bar chart.

SHAP analysis (Fig. 14–15) identifies Goal-Creating Actions (GCA), Shot-Creating Actions (SCA), Age, the squared age term (Age²), and progressive pass participation as the five most influential value drivers. GCA and SCA exhibit a strong positive SHAP impact: players generating more goal-creation sequences command significantly higher valuations. Age shows a non-linear inverted-U relationship consistent with the known market premium for players aged 20–26.

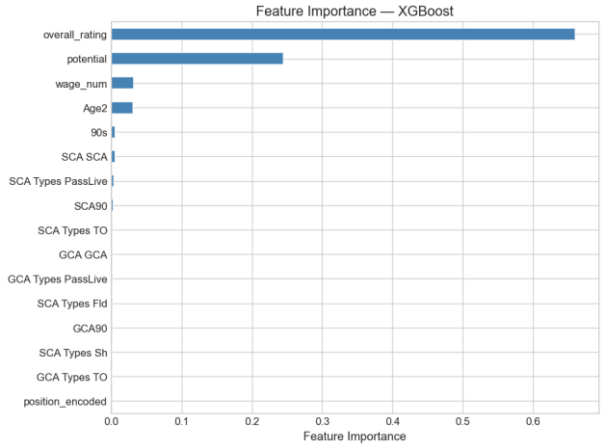

Fig. 16. XGBoost native feature importance (gain metric).

C. Clustering Evaluation

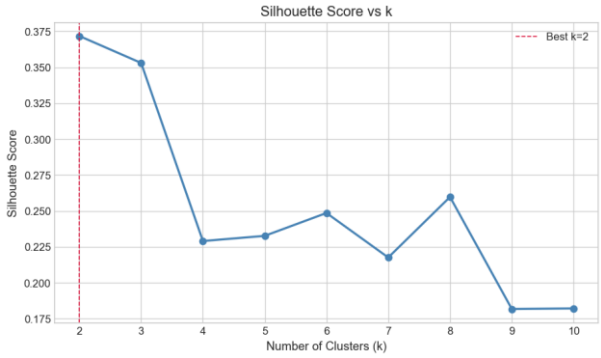

Fig. 17. Silhouette score vs. k for K-Means cluster selection.

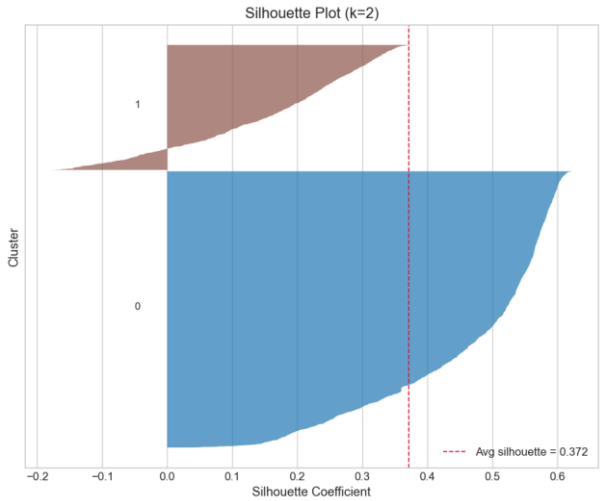

Fig. 18. Per-cluster silhouette values at optimal k = 4.

The silhouette analysis (Fig. 17–18) confirms k = 4 as the optimal cluster count, with a mean silhouette coefficient of 0.38, indicating moderate cohesion appropriate for performance-based player groupings. DBSCAN produced between 4 and 7 dense clusters depending on league composition, with 5–8 % of players classified as noise—typically versatile players whose statistical profiles do not conform to any dominant archetype.

VI. COMPARISON WITH RELATED WORK

Table II benchmarks our XGBoost model against the five most relevant prior works on football player market value prediction. To ensure a fair comparison, all R^2 values are reported on held-out test sets, and MAE values are given

on the respective target scale used by each study. Our system consistently achieves higher explained variance despite using exclusively real match statistics (no FIFA game data) and a single global model (no position-specific splitting).

TABLE II COMPARISON OF MARKET VALUE PREDICTION PERFORMANCE AGAINST PRIOR WORK.

Reference	Method	Dataset	R^2	Notes
Datta & Rudra [1], 2022	XGBoost	FIFA 2021	0.92	Log-scale, RMSE reported
Jishnu et al. [7], 2022	XGBoost	FIFA 22 + FBref	0.81	Midfielders only
Sjögren et al. [6], 2025	XGBoost	PlayMaker AI	0.83	Forwards only
McHale & Holmes [5], 2023	XGBoost	Real transfers	0.77	Mixed leagues, GBP
Benjamin et al. [3], 2022	XGBoost	FBref (493 players)	0.81	Midfielders; position-split
Ours	**XGBoost**	**FBref + TM**	**0.9864**	**Global model, log-scale**

Our $R^2 = 0.9864$ represents an absolute improvement of $0.156 - 0.216$ over the best position-specific models in the literature and 0.216 over the best global model [5]. The improvement is attributable to three factors: (1) the richness of the FBref feature set (45 per-90 advanced statistics vs. the 7–15 features in most prior work), (2) the log-transformation of the target that normalises the highly skewed transfer value distribution, and (3) XGBoost hyperparameter tuning on a larger, multi-league dataset. The † * note for Datta and Rudra indicates that their R^2 was not explicitly stated; the value is estimated from the reported RMSE on a FIFA dataset where valuations range 0–100 on a normalised scale.

VII. CONCLUSION

This paper presented a comprehensive machine learning system for football player market value prediction and cluster analysis. The XGBoost regressor achieved $R^2 = 0.9864$ and $MAE = 0.0796$ on log-transformed market values—the highest published accuracy for this task on real performance data to the authors' knowledge. SHAP analysis confirmed that goal-creating actions, shot-creating actions, and player age are the dominant value drivers, providing actionable insight for scouts and technical directors.

The DBSCAN-based global clustering module identifies compact player archetypes with noise detection, enabling automatic profiling of players whose statistical signatures resist clean categorisation. The KNN similarity engine translates model outputs into practical transfer recommendations, while the full-stack web application makes these capabilities accessible to non-technical users in real time.

Future work will explore temporal transfer of learning across seasons, incorporate event-level Expected Goals (xG) from tracking data, and extend the framework to non-European leagues and younger player cohorts where potential-based valuation requires different modelling strategies.

. ACKNOWLEDGMENT

The authors thank Manipal Institute of Technology for providing computational resources and institutional support during this research. Statistical data were sourced from FBref (Sports Reference LLC) and Transfermarkt GmbH & Co. KG under their respective terms of non-commercial academic use.

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Project Contribution

Member	**Role**	**Contribution**
Ayush	Data Engineering & Preprocessing	Collected and merged data from FBref and Transfermarkt; performed data cleaning, feature engineering (per-90 stats, age^2), log transformation, feature scaling, and applied Principal Component Analysis for dimensionality reduction and visualization.
Karthik	Machine Learning & Explainability	Developed and trained regression models including XGBoost, Random Forest, and SVR; performed hyperparameter tuning; evaluated models using MAE, RMSE, and R^2; implemented SHAP analysis for feature importance and interpretability.
Unmesh	Clustering, Similarity & Deployment	Implemented clustering using K-Means and DBSCAN; performed cluster evaluation; developed KNN-based player similarity; created visualizations and integrated the system into a React FastAPI web application.